

A System-Oriented Design and Simulation Framework for a University Student Mental Health Support Platform

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Abstract—University students frequently experience emotional distress, yet institutional mental health services are often perceived as difficult to access, poorly visible, or insufficiently trusted. This paper proposes a system-oriented mental health support platform that combines anonymous intake, tiered support pathways, peer counselor matching, professional referral, privacy governance, and continuous monitoring as part of a broader university wellbeing ecosystem. Survey evidence and simulation results suggest that the proposed design can reduce access barriers under normal conditions, while stress-period demand, peer counselor availability, and user trust remain the main operational variables that must be controlled before pilot deployment.

Index Terms—mental health support, systems analysis, system design, discrete-event simulation, privacy by design, university wellbeing, peer support

I. INTRODUCTION

University mental health support is not only a clinical or technological issue; it is a socio-technical problem shaped by student behavior, institutional visibility, trust, privacy, professional capacity, and the availability of first-contact support channels. In the analyzed university context, the project survey showed that many students experience emotional or academic stress but do not necessarily use existing services, which indicates a gap between need and action rather than a simple lack of institutional resources. This gap is particularly relevant in academic environments where students may avoid formal support because of stigma, uncertainty about access routes, fear of identification, or low confidence in available channels.

The project addresses this problem through the design of a Student Mental Health Support Platform understood as one component of a larger university wellbeing system. Instead of replacing professional counseling or performing clinical diagnosis, the platform is intended to provide orientation, anonymous first contact, resource access, peer support, referral to professional counselors, and operational monitoring. This interpretation is consistent with systems engineering principles because it begins with actors, processes, responsibilities, decision points, and information flows before defining software modules.

The main contribution of this work is the integration of four stages of systems analysis and design into a compact validation framework. First, primary survey data was used to identify barriers, preferences, and support needs. Second, a system architecture was defined around stakeholders, support pathways, privacy governance, and escalation processes. Third, quality assurance, risk management, and project management mechanisms were added to prepare the design for implementation. Finally, computational simulations were used to test whether the design behaves acceptably under baseline, academic stress, counselor shortage, and trust-collapse scenarios.

This paper follows a concise IEEE-style structure. Section II presents the methods and materials, including data collection, system architecture, quality controls, and simulation design. Section III presents the main results and discusses their implications for design validation. Section IV concludes the work and identifies future actions required before institutional deployment. Related concerns in digital mental health, privacy, accessibility, quality management, risk management, project governance, and simulation-based validation are supported by established references and standards such as WHO guidance on digital mental health, WCAG accessibility principles, ISO 9001, ISO 31000, PMBOK, and discrete-event simulation practices [1], [2], [3], [4], [5], [6].

II. METHODS AND MATERIALS

A. Empirical Basis and Problem Characterization

The first stage of the project consisted of a structured survey applied to 25 university students. Although the sample is limited, the survey provided useful early evidence for design decisions. The most relevant finding is the coexistence of high emotional need and low use of formal support channels. Table I summarizes the main indicators used as design inputs.

The survey results were interpreted cautiously because they include both objective counts and subjective perceptions. However, they were sufficient to justify the main design direction: the system should not only provide a digital interface, but also reduce first-contact friction, clarify support pathways, preserve anonymity, and coordinate different levels of care.

TABLE I: Survey Indicators Used as Design Inputs

Indicator	Result	Design meaning
Anxiety or emotional problems during university life	88%	High latent need for accessible support.
Frequent or constant academic stress	72%	Demand is expected to increase during academic pressure periods.
Never sought mental health help	68%	There is an access, trust, and awareness gap.
Rated institutional access as easy or very easy	0%	Existing pathways are not sufficiently visible or simple.
Preferred anonymous support	64%	Privacy is a behavioral requirement for adoption.
Open to using a digital platform	76%	A platform has adoption potential if it is trustworthy.
Requested professional, peer, and educational support together	68%	The system must offer multiple support pathways.

Figure 1. System Context and Stakeholder Architecture

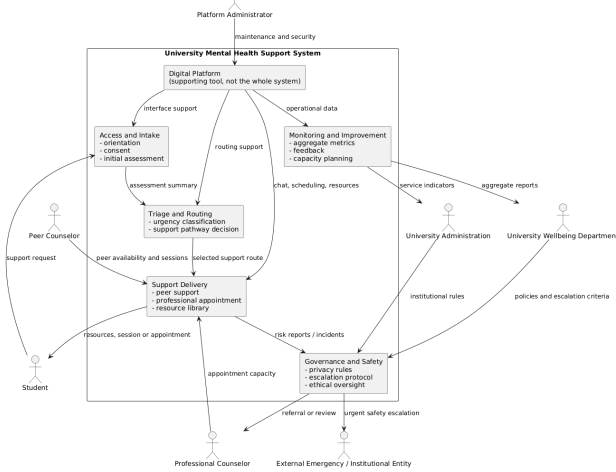


Fig. 1: System context and stakeholder architecture of the Student Mental Health Support Platform.

B. System-Oriented Design

The proposed solution was modeled as a socio-technical system. The main actors are students, peer counselors, professional counselors, the university wellbeing department, platform administrators, and external emergency or institutional entities. The core processes include intake, informed consent, needs assessment, triage, peer assignment, professional referral, escalation, feedback collection, and continuous improvement.

Figure 1 presents the system context and stakeholder architecture developed during the design phase. It shows that the platform is not the complete system, but the technological coordination layer that supports human and institutional processes.

At the architectural level, the design separates three levels of responsibility. The first level is the university mental health support ecosystem, which includes students, peer networks, professional services, wellbeing governance, and emergency

Figure 2. System-Oriented Robust Architecture

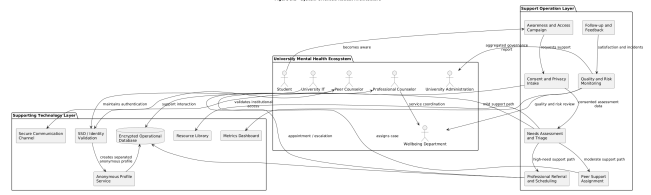


Fig. 2: Robust system architecture showing ecosystem, operational process, and supporting technology layers.

protocols. The second level is the operational process layer, where support requests are transformed into decisions such as self-help routing, peer matching, appointment scheduling, or escalation. The third level is the supporting technology layer, which includes secure authentication, anonymous profiles, resource access, scheduling, communication, monitoring dashboards, and data infrastructure.

The design uses a tiered support model. Low-intensity cases can be guided toward curated educational resources, moderate support requests can be routed to trained peer counselors, and higher-need cases can be referred to professional counseling services. Escalation rules are necessary because peer counselors are not clinical professionals and should not handle high-risk cases independently. Privacy governance is treated as a system-level control through consent, data minimization, role-based access, separation between institutional identity and anonymous support profile, and retention limits.

C. Quality, Risk, and Implementation Planning

The robust design stage added quality assurance and risk management mechanisms. The quality framework considered availability, usability, accessibility, security, maintainability, response time, referral effectiveness, adoption, and peer workload control. Risk analysis focused on operational risks such as peer counselor shortage, social risks such as low adoption or loss of trust, ethical risks such as inappropriate peer interactions, and technical risks such as authentication failure or de-anonymization.

Figure 2 presents the robust architecture produced during the project management and quality assurance phase. It connects the support ecosystem, operational controls, governance mechanisms, and supporting technology.

The project management plan defined phased implementation: architecture refinement, quality and risk planning, project management integration, pilot preparation, and final documentation. For a future pilot, the minimum readiness conditions are a validated privacy model, trained peer counselors, escalation rules, monitoring dashboards, acceptance criteria, and a clear communication plan with the university wellbeing department.

D. Simulation Model and Scenarios

The validation stage used computational simulation to test whether the design remains coherent under different operational conditions. The model does not represent a full clinical service or production application; it abstracts the parts required

TABLE II: Process-Oriented Simulation Results by Scenario

Scenario	Avg. wait (min)	Util. (%)	Dropouts (n/%)	Served
Baseline	0.00	3.53	0 / 0.00%	500
Midterm/Finals Stress	0.00	8.62	18 / 3.60%	482
Counselor Shortage	0.00	10.58	0 / 0.00%	500
Trust Collapse	0.00	4.72	0 / 0.00%	500

for early design validation: student entry, trust-sensitive registration, profile creation, peer counselor assignment, counselor workload, response time, and dropout behavior. The main simulation approach was discrete-event simulation because student arrivals, support requests, queueing, assignments, and dropouts occur as discrete events.

The main parameters were a population of 500 simulated students, 30 peer counselors, a peak multiplier of 2.5 during academic stress periods, an average response time target of 5 minutes, a burnout threshold of 20 sessions, and a baseline trust level of 0.85. Four scenarios were tested: Baseline, Midterm/Finals Stress, Counselor Shortage, and Trust Collapse. These scenarios were selected because they correspond directly to risks identified in the previous workshops: ordinary demand, academic demand surge, limited human capacity, and erosion of user trust.

III. RESULTS AND DISCUSSION

A. Design Validation from Survey Evidence

The survey confirmed that the most important system problem is not only the existence of emotional distress, but the gap between need and use of available support. The combination of 88% emotional difficulty, 68% non-help-seeking, and 0% easy-access perception justifies a platform focused on orientation and first contact. The 64% anonymity preference supports the decision to separate institutional authentication from anonymous support interactions. Finally, the 68% preference for comprehensive support validates the tiered model because students do not appear to require only one support channel.

These findings support the system-oriented architecture. A purely software-centered design would focus on login, chat, and database modules, but the survey suggests that the real challenge is coordination: students must understand where to go, feel safe enough to begin, receive an appropriate support level, and be referred when the case exceeds peer support boundaries.

B. Process-Oriented Simulation Results

The process-oriented simulation evaluated waiting behavior, counselor utilization, dropout rate, and service capacity. Table II summarizes the four scenarios. Under normal conditions, the system operates with a large safety margin. During midterm stress, the main failure is not sustained overload but short demand bursts that generate 18 dropouts. Under counselor shortage, utilization increases but the model still serves all students. Under trust collapse, the technical system remains available, but low trust reduces meaningful use.

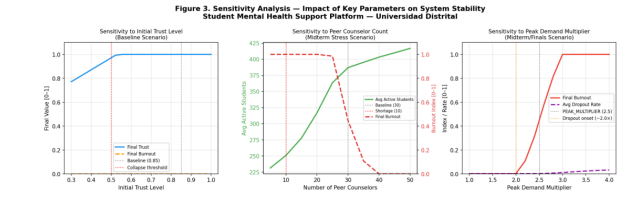


Fig. 3: Sensitivity analysis of trust, peer counselor capacity, and peak demand multiplier on system stability.

The Midterm/Finals Stress result is the most relevant operational finding. Average wait time remains near zero, but the dropout count reveals transient queue spikes. Therefore, average wait time alone is not enough as a safety indicator. A real-time queue monitor and proactive waiting messages are recommended because they would address short bursts more directly than simply adding more counselors.

C. Behavior-Oriented Simulation and Sensitivity

The behavior-oriented simulation explored trust, adoption, dropout, burnout, and recovery patterns across semester-level scenarios. The main insight is that trust behaves like a structural variable rather than a cosmetic feature. When trust declines, the system may become underused even if technical resources remain available. This validates the privacy-by-design approach and the need for transparent communication, consent, and incident response protocols.

Figure 3 presents the sensitivity analysis obtained from the simulation stage. It shows how initial trust, peer counselor capacity, and peak demand influence system stability. This figure is central to the discussion because it demonstrates non-linear behavior: small changes in trust or capacity can produce disproportionate effects on dropout, burnout, or recovery.

The simulations validate several design decisions. First, the system needs an academic-calendar-aware demand strategy because stress periods generate short but important demand peaks. Second, peer counselor capacity must be monitored not only by headcount but also by workload and burnout thresholds. Third, privacy governance is operationally relevant because trust affects adoption and effectiveness. Fourth, the platform should be piloted before full deployment because the current model simplifies session difficulty, peer fatigue, student retry behavior, and professional counselor workflows.

D. Quality and Acceptance Implications

The results suggest that the design is feasible as an early support and referral system if it is implemented with strong governance. The most important acceptance metrics for a pilot are not only technical uptime or response time, but also intake completion, referral accuracy, peer workload, adoption rate, dropout rate, privacy incidents, and user trust. A future pilot should compare real user behavior with the simulated thresholds and adjust the matching policy, peer recruitment strategy, and communication protocols accordingly.

IV. CONCLUSIONS

This project designed and validated a Student Mental Health Support Platform as part of a broader university wellbeing ecosystem. The work was successful because it translated survey evidence into a system architecture that addresses access barriers, anonymity, multi-channel support, professional referral, risk management, and continuous improvement. The main value of the proposal is not the platform as an isolated application, but its role as a coordination mechanism between students, peer support networks, professional counseling services, institutional governance, and privacy controls.

The simulation results support the general feasibility of the design under baseline conditions and reveal the most important vulnerabilities before implementation. Midterm and finals periods require surge monitoring, counselor shortage requires capacity planning, and trust collapse requires privacy governance and transparent institutional communication. Future work should conduct a controlled pilot, collect real operational data, validate the escalation process with the wellbeing

department, and refine the simulation model using observed arrival rates, session durations, peer fatigue indicators, and professional referral outcomes.

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